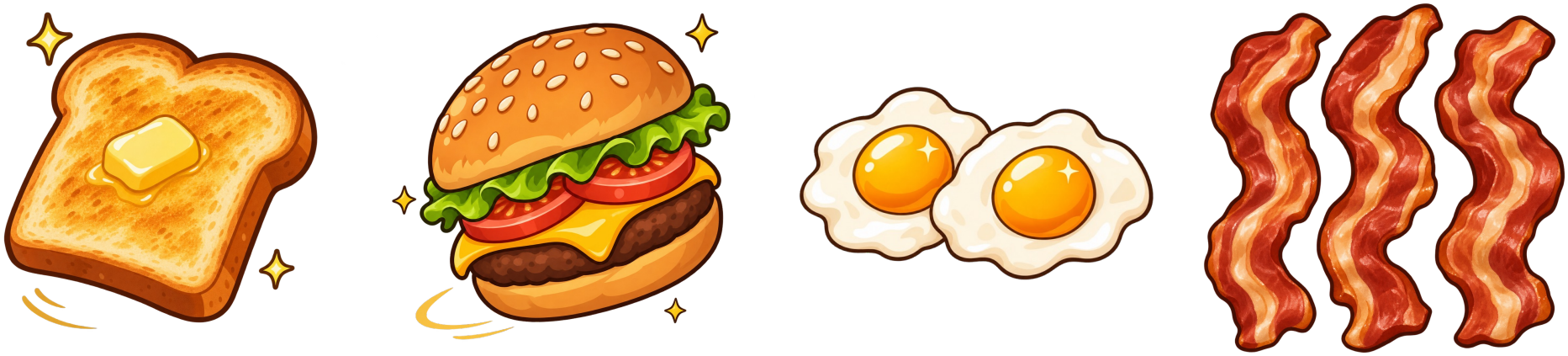
Meet the adult morphs

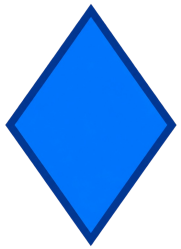
(observable covariate)

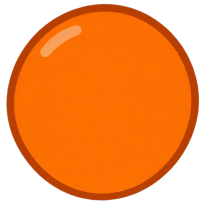


Meet the juvenile morphs

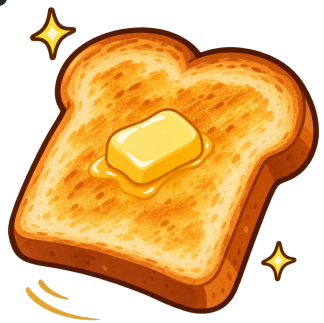
(observable covariate)







The probability that a particular, *sampled* space toast, and a particular *sampled* green shamrock are a POP depends on....

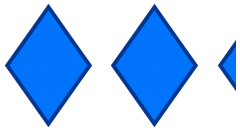


1) the expected number of shamrocks produced by a space toast that would be around at the time of sampling of the juvenile....

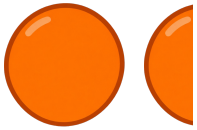


3.6

(Numerator)



2.2

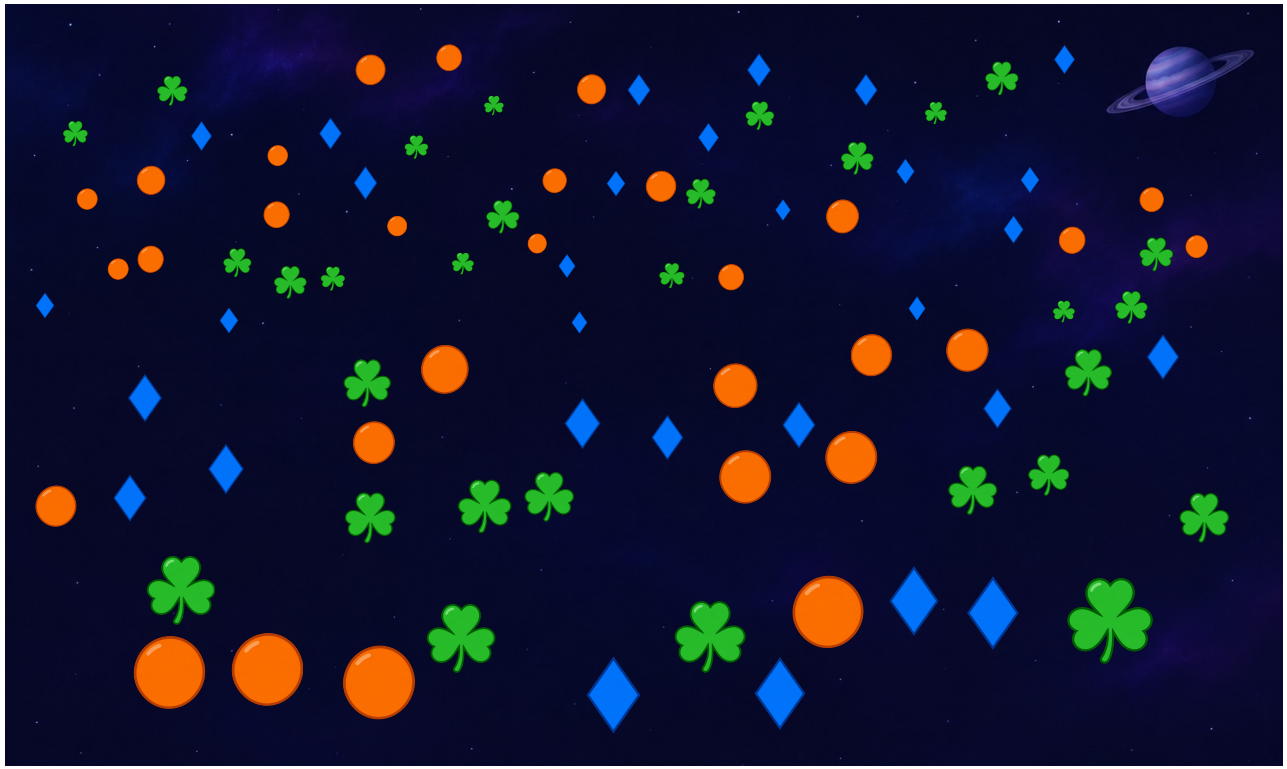


1.4

The probability that a particular, *sampled* space toast, and a particular *sampled* green shamrock are a POP depends on....

2) ...**AND** the expected total number of shamrock morphs, around at the time of juvenile sampling, produced by the whole population of adults.

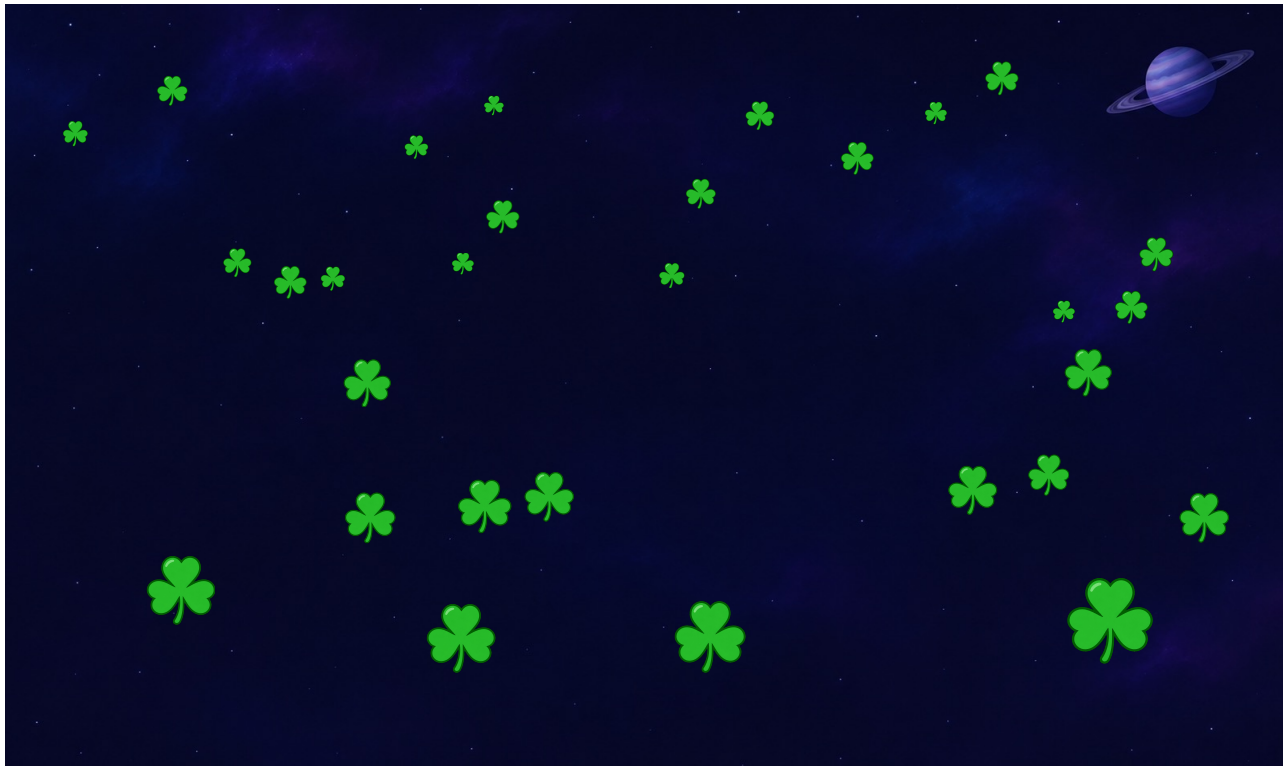
(Denominator)



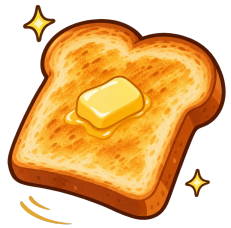
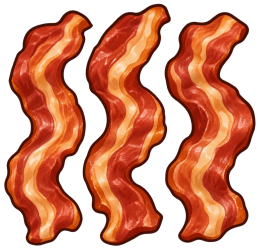
The probability that a particular, *sampled* space toast, and a particular *sampled* green shamrock are a POP depends on....

2) ...**AND** the expected total number of shamrock morphs, around at the time of juvenile sampling, produced by the whole population of adults.

(Denominator)



That denominator depends on the number of adults of different types and their expected reproductive outputs

N_{st}	N_{cb}	N_{ssu}	N_{tsb}
			
3.6	**1.2**	**2.4**	**0.8**
2.2	3.4	1.6	2.8
1.4	2.6	3.2	1.8

POP probability is expected output of green shamrocks of focal mother divided by expected output of green shamrocks from the total poulation

$$P(\text{POP} \mid \text{toast}, \text{shamrock}) = e_{st} / (\Sigma_m N_m e_m)$$

e_m = expected shamrocks per adult of morph m